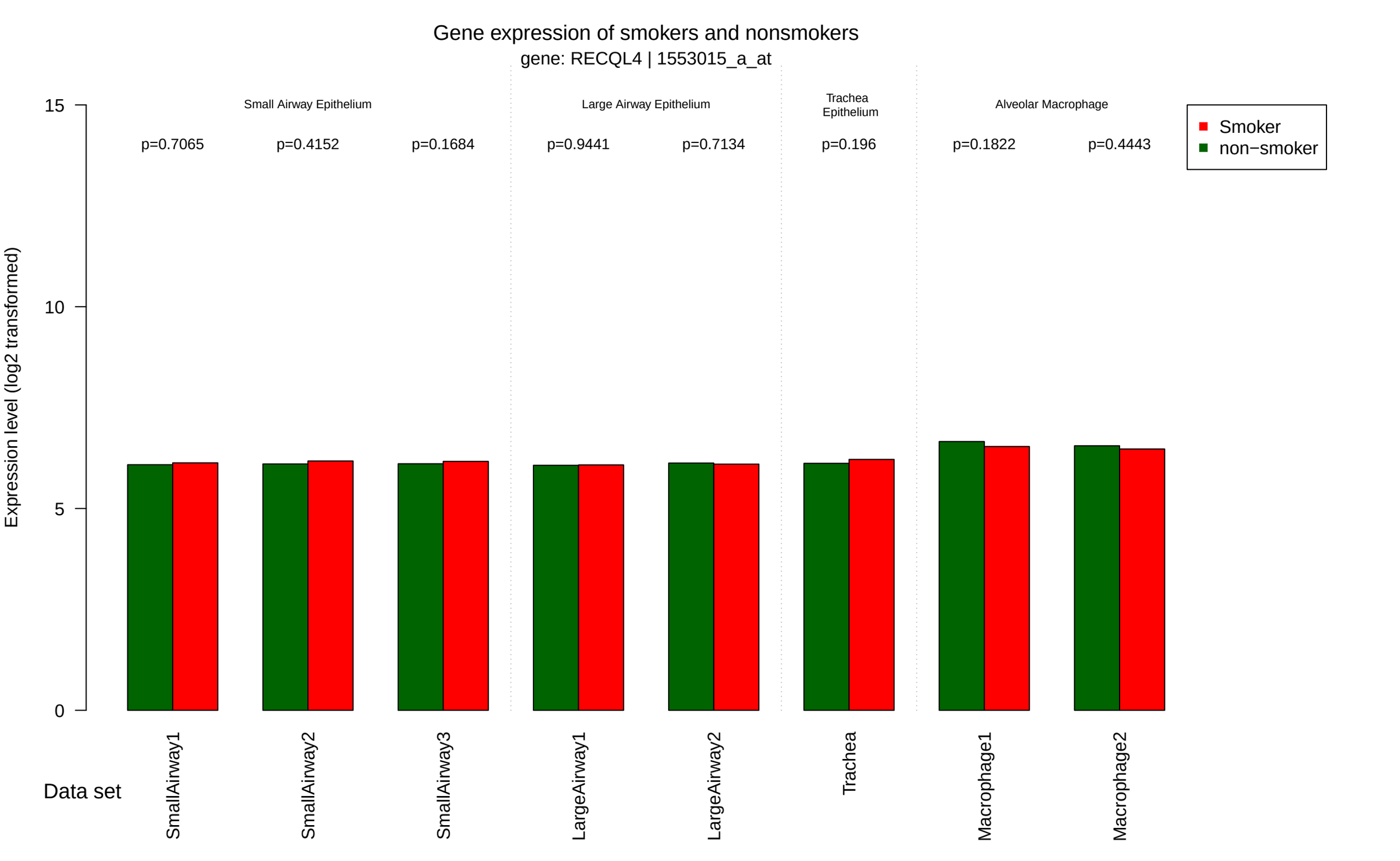
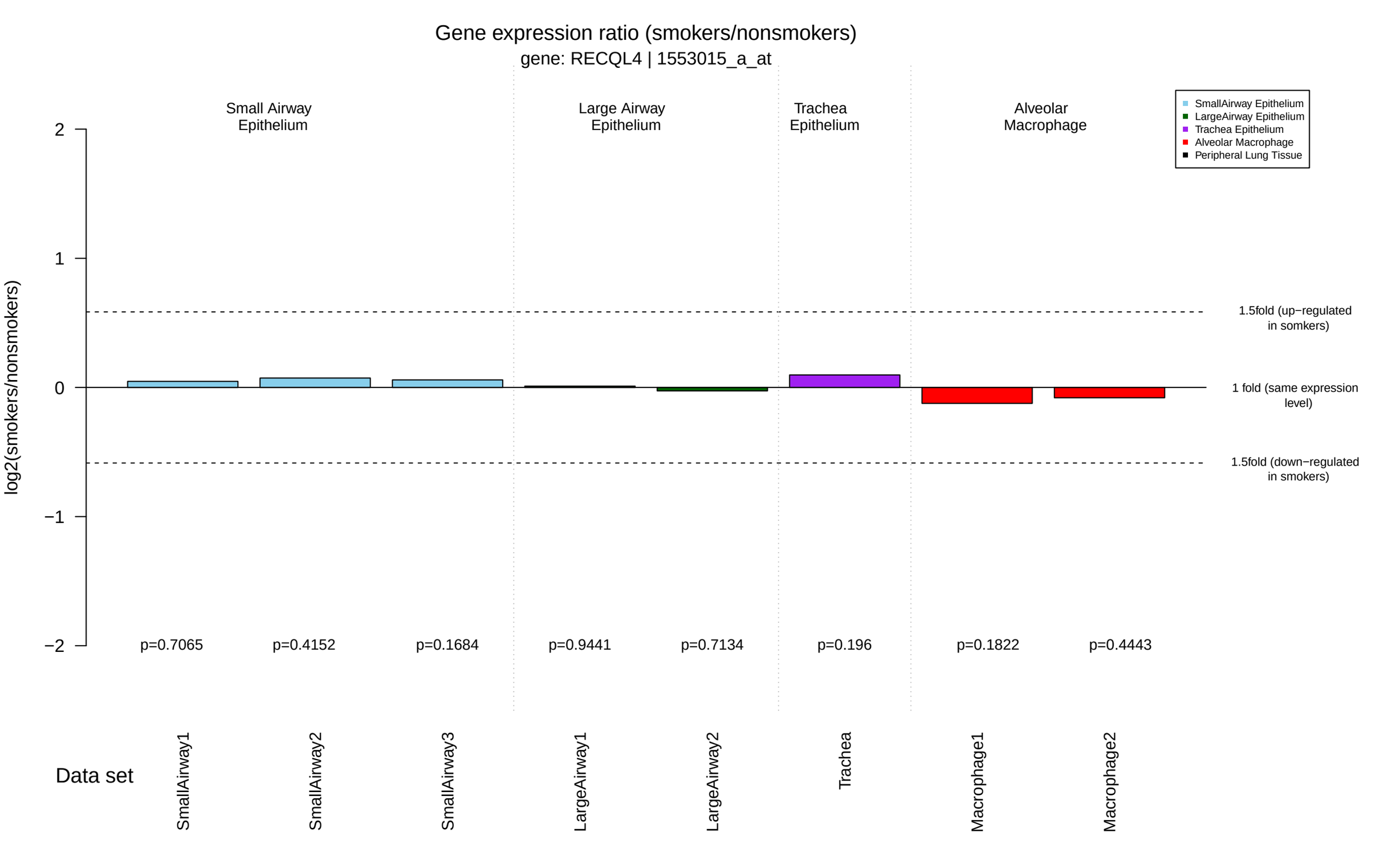
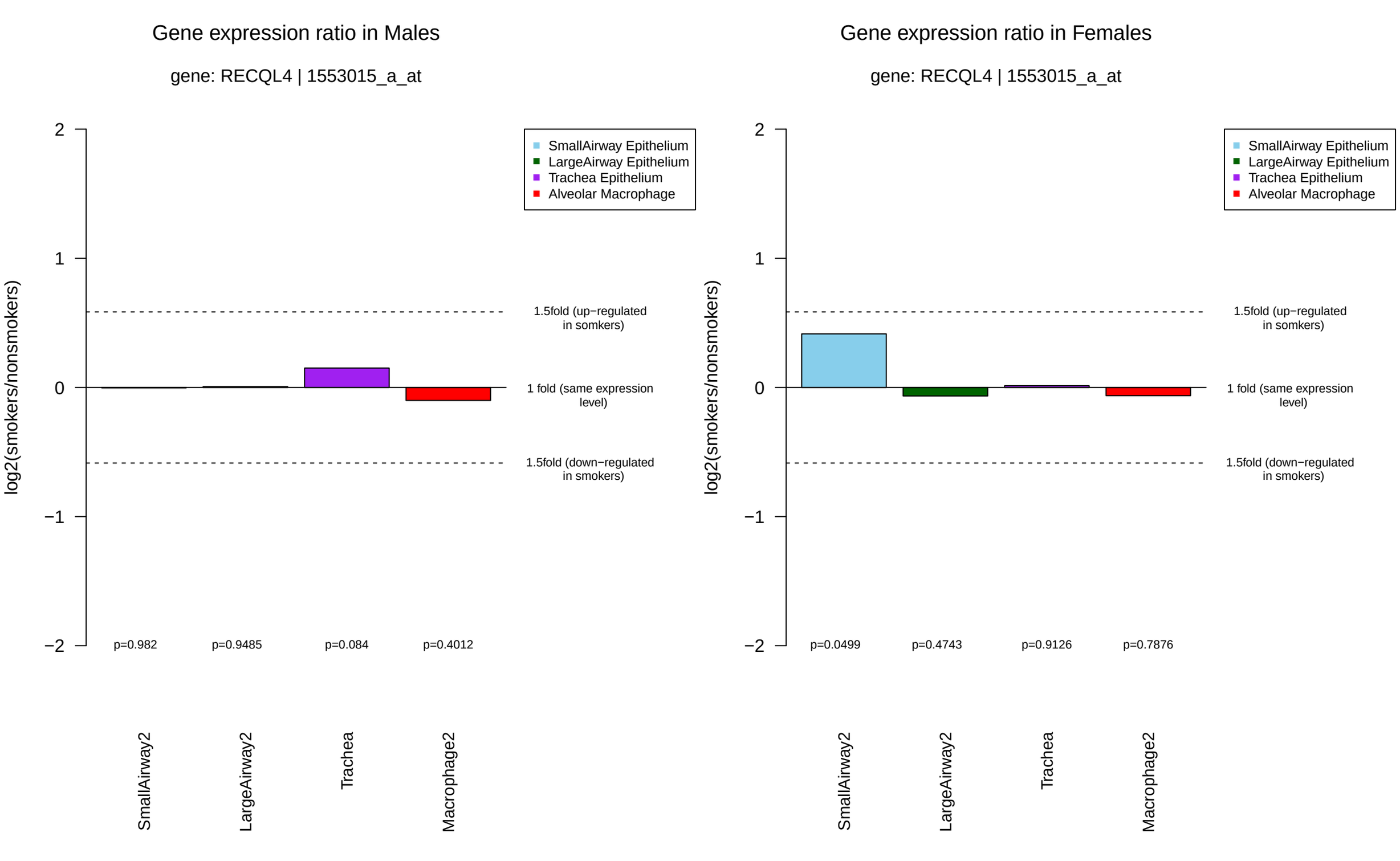
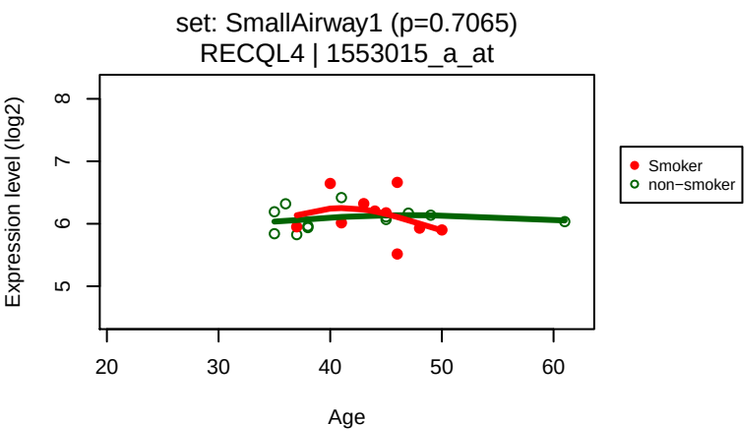
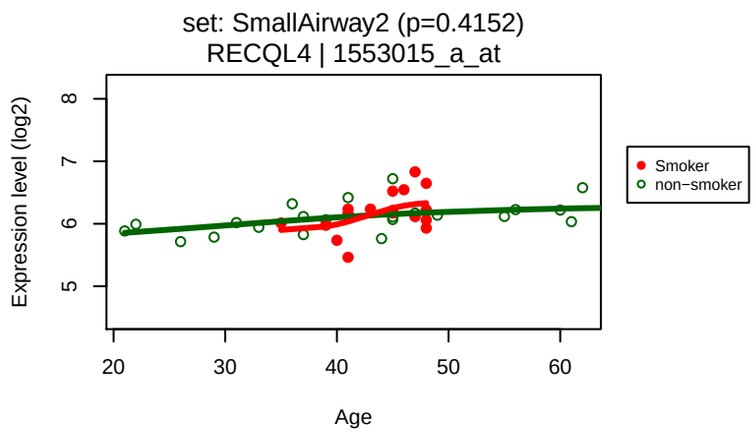
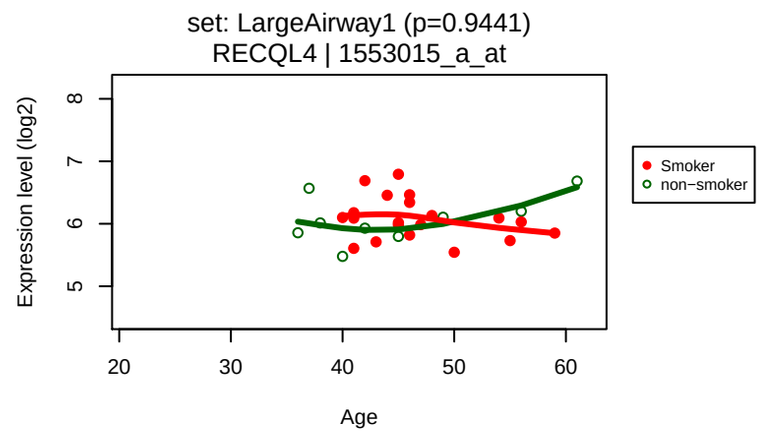

Gene expression of smokers and nonsmokers
gene: RECQL4 | 1553015_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7065
p=0.4152
p=0.1684
p=0.9441
p=0.7134
p=0.196
p=0.1822
p=0.4443
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

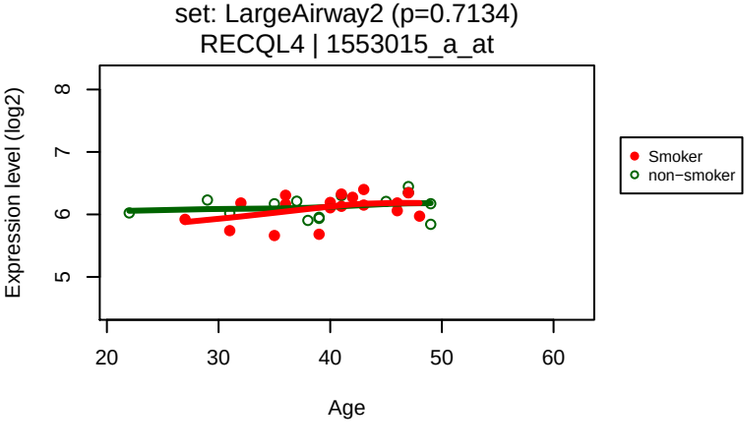

Gene expression ratio (smokers/nonsmokers)
gene: RECQL4 | 1553015_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7065
p=0.4152
p=0.1684
p=0.9441
p=0.7134
p=0.196
p=0.1822
p=0.4443
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

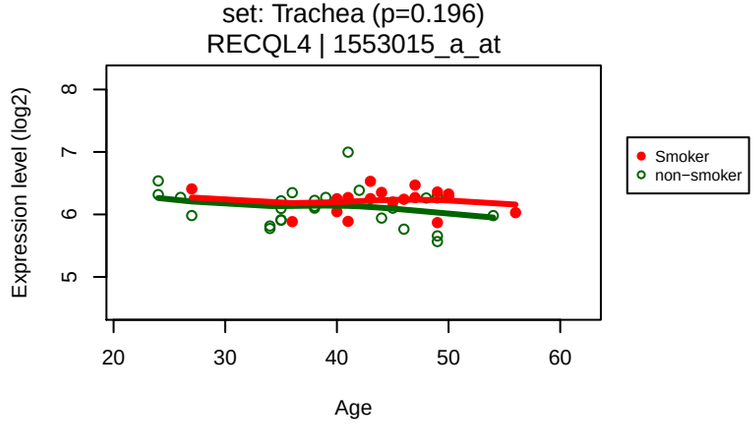

Gene expression ratio in Males
gene: RECQL4 | 1553015_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.982
p=0.9485
p=0.084
p=0.4012
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RECQL4 | 1553015_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0499
p=0.4743
p=0.9126
p=0.7876
SmallAirway2
LargeAirway2
Trachea
Macrophage2

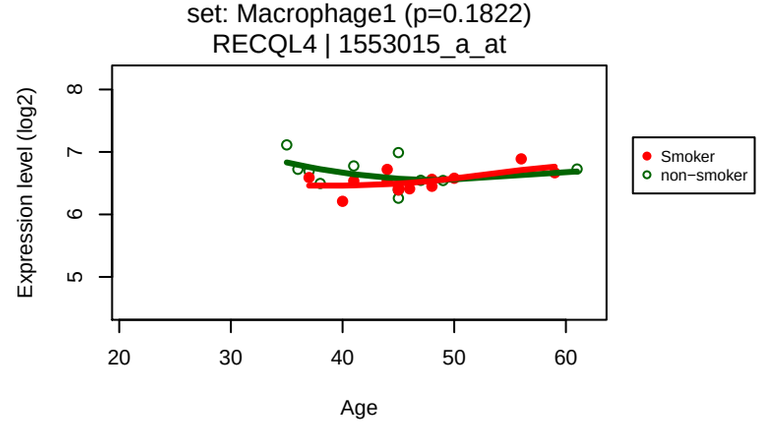

set: SmallAirway1 (p=0.7065)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker

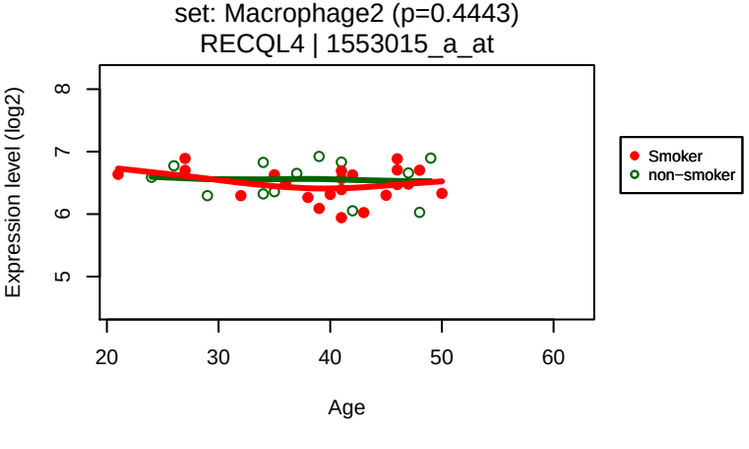

set: SmallAirway2 (p=0.4152)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker

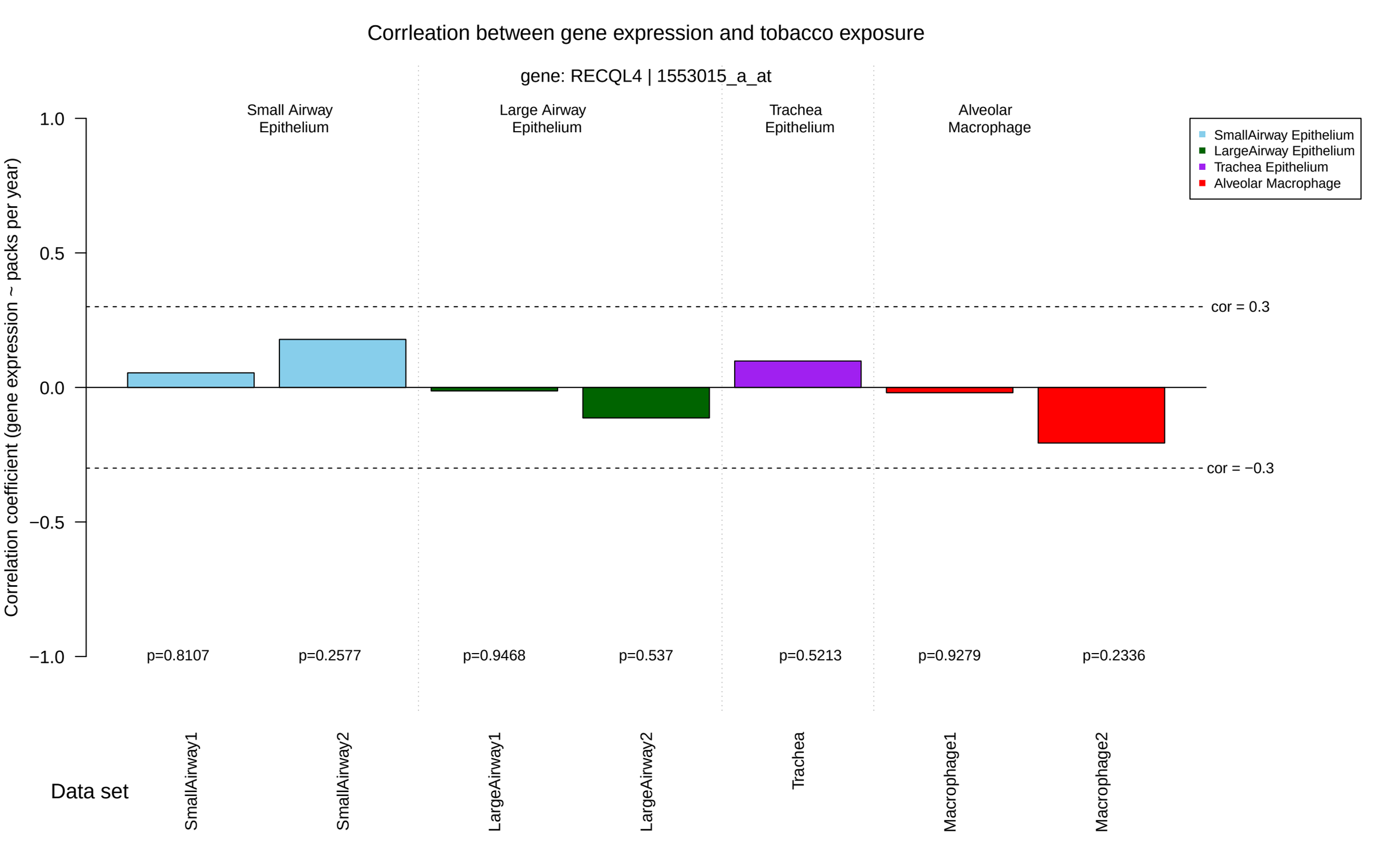

set: LargeAirway1 (p=0.9441)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7134)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.196)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1822)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4443)
RECQL4 | 1553015_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RECQL4 | 1553015_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8107
p=0.2577
p=0.9468
p=0.537
p=0.5213
p=0.9279
p=0.2336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2